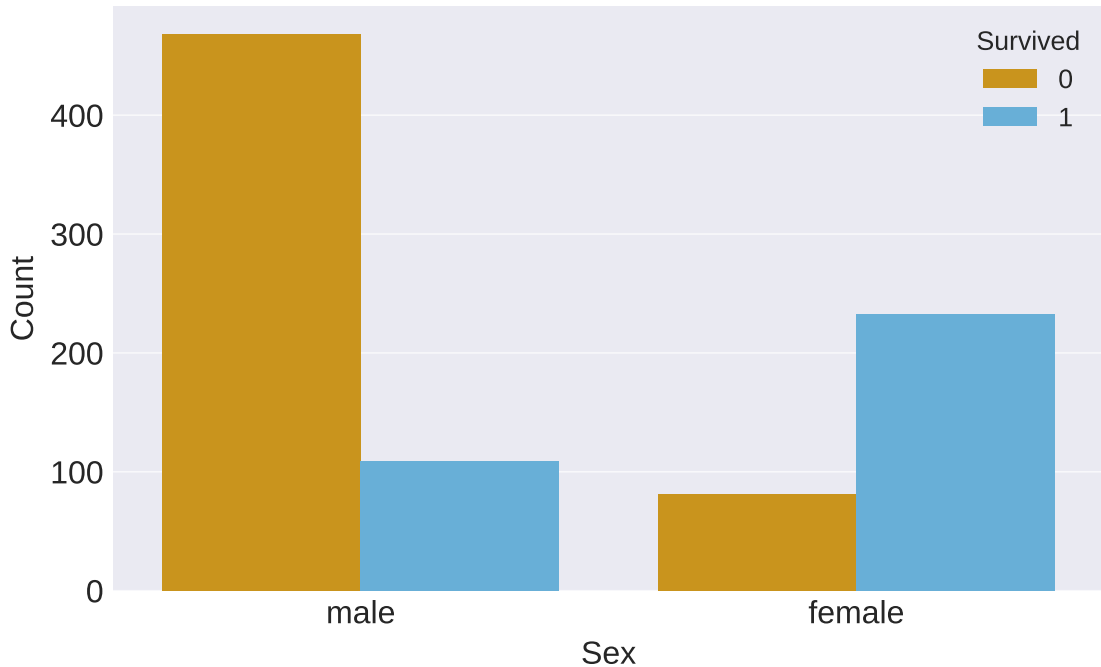


Titanic Dataset - Exploratory Data Analysis (EDA)

Deliverables: Jupyter Notebook + PDF report of findings

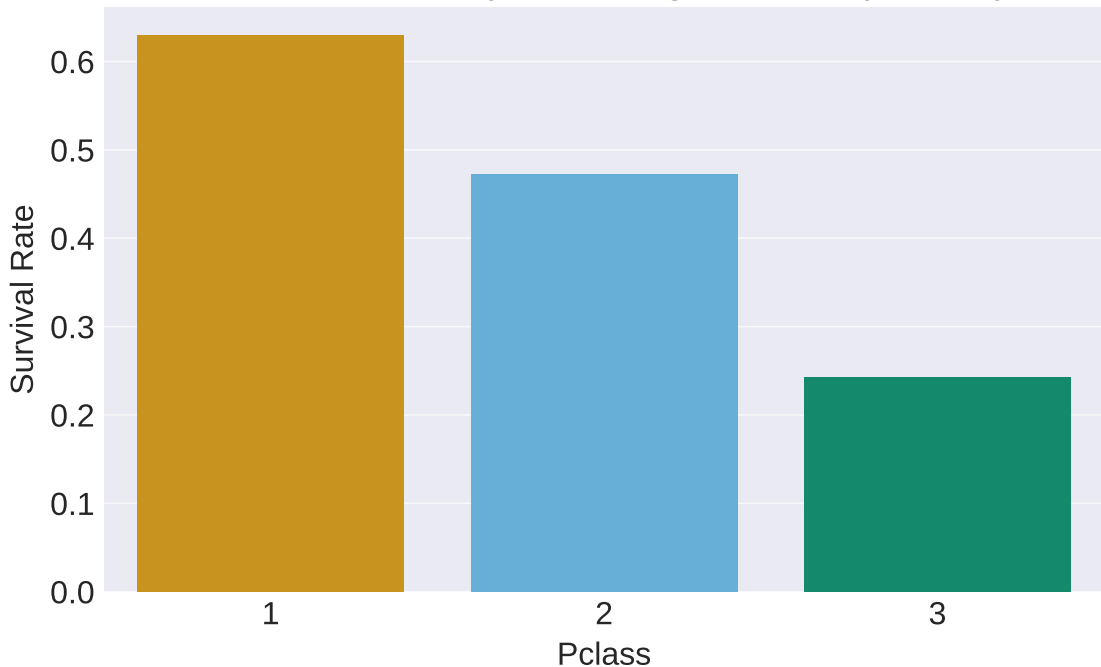
Survival Count by Sex



Survival Count by Sex

Count of survivors vs non-survivors grouped by sex (male/female).

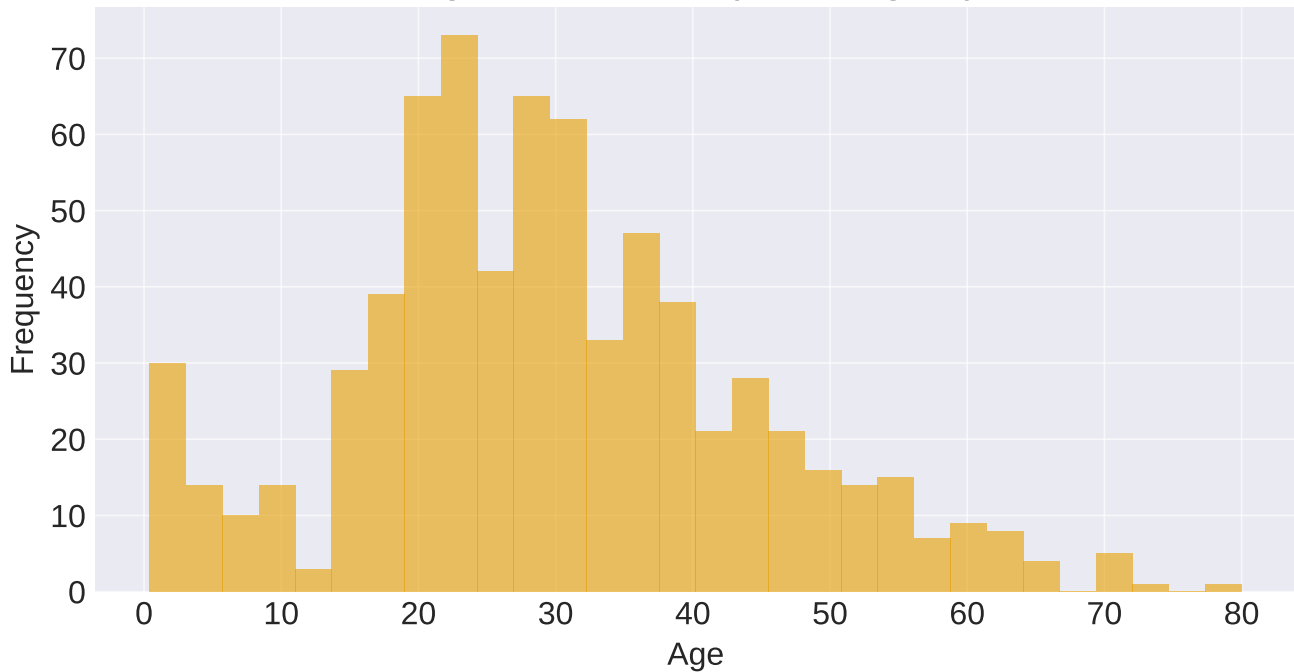
Survival Rate by Passenger Class (Pclass)



Survival Rate by Pclass

Average survival rate for each passenger class (1,2,3).

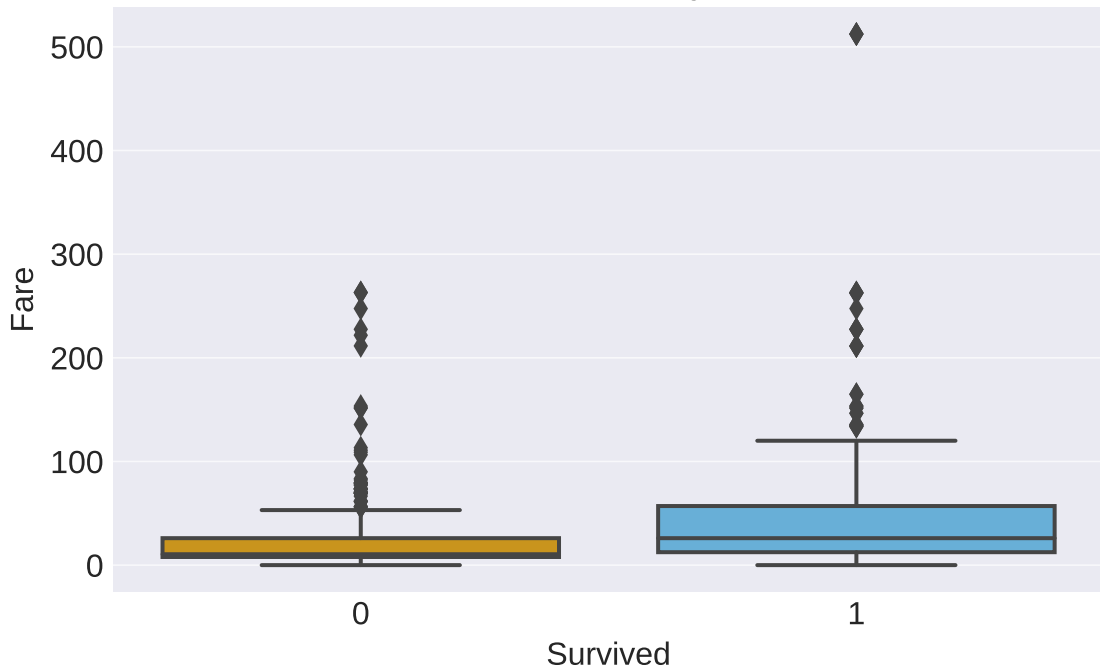
Age Distribution (Passengers)



Age Distribution

Distribution of passenger ages (missing ages filled for grouping where needed).

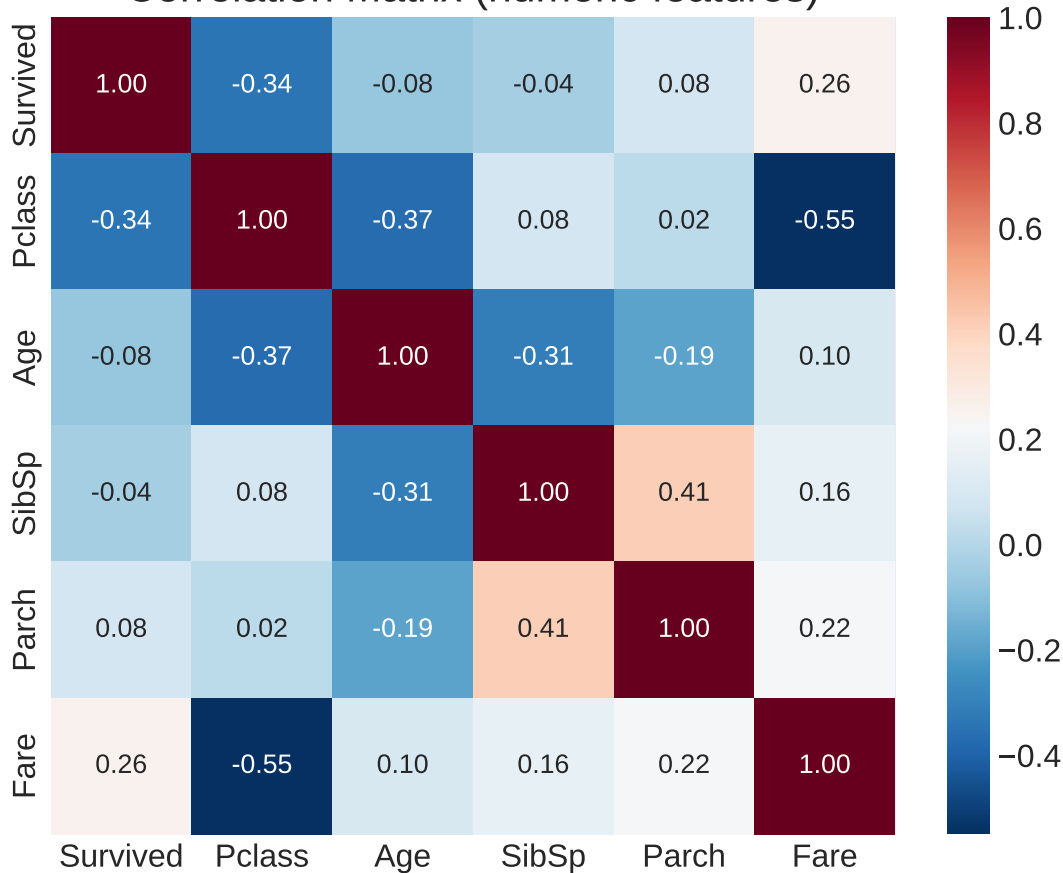
Fare Distribution by Survival



Fare Distribution by Survival

Boxplot of fare paid by survivors vs non-survivors.

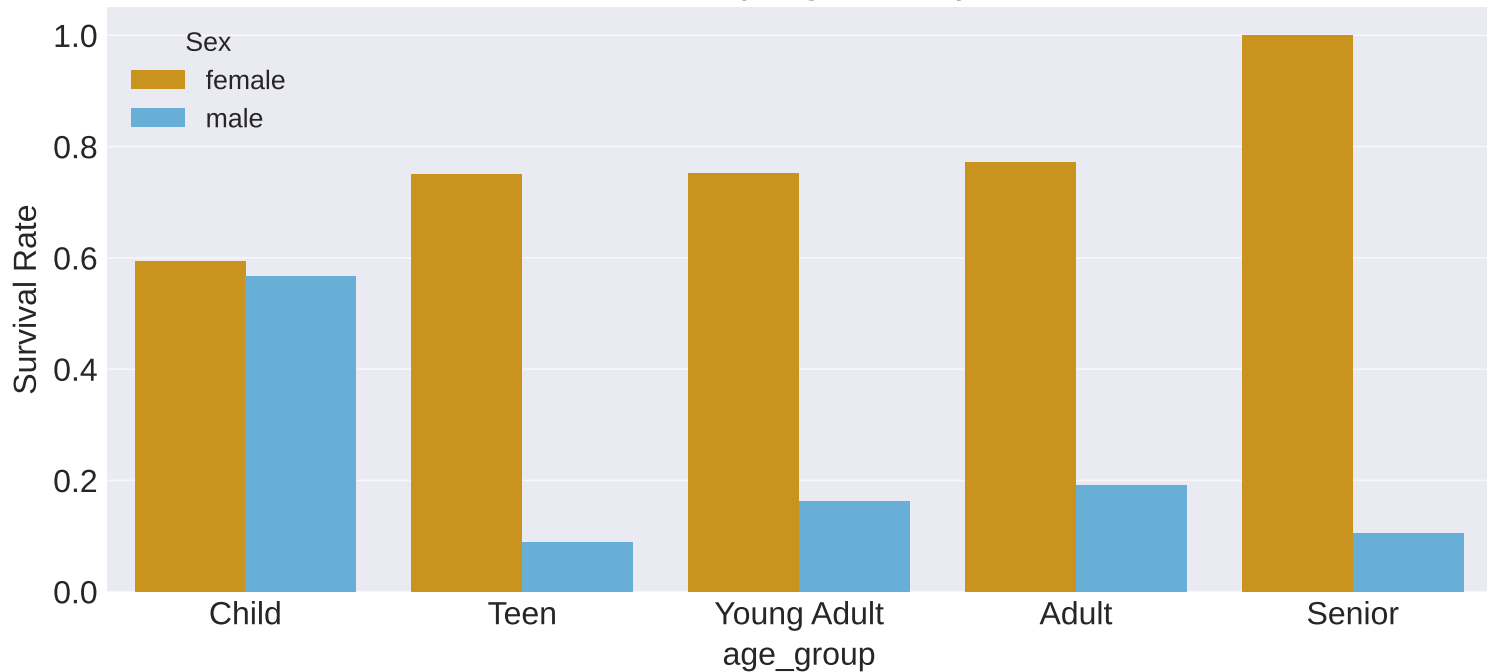
Correlation Matrix (numeric features)



Correlation Matrix

Correlation heatmap of numeric features (Survived, Age, Fare, etc.).

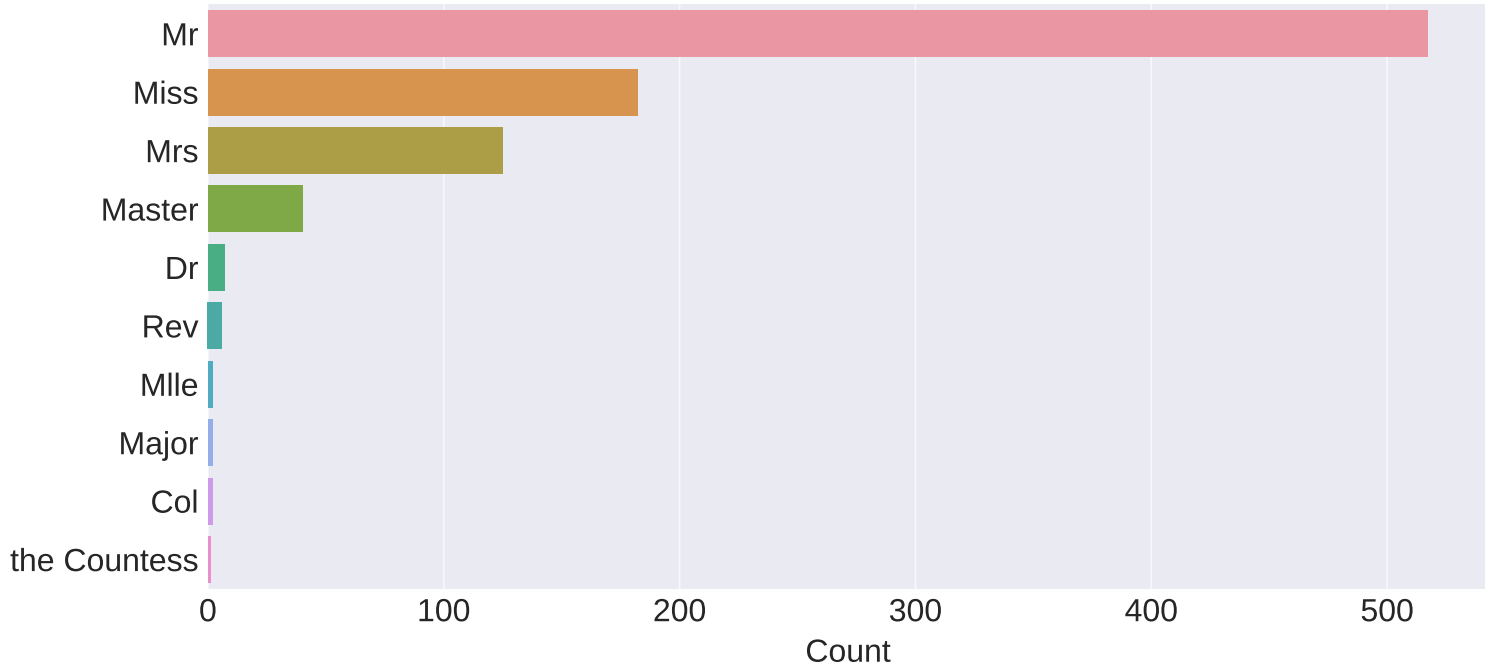
Survival Rate by Age Group and Sex



Survival Rate by Age Group and Sex

Survival by age group and sex — shows strong female advantage.

Top 10 Titles in Names (proxy for social title)



Top 10 Titles in Names

Most common name titles (Mr, Mrs, Miss) serving as a proxy for social titles.

Summary of Key Findings:

1. Sex is a strong predictor of survival: females have a much higher survival rate than males.
2. Passenger class (Pclass) matters: higher classes have higher survival rates.
3. Age distribution shows a mix of ages; children and young adults show different survival behaviors.
4. Fare tends to be higher for survivors; higher fare (proxy for higher class) linked to better survival.
5. Correlations: 'Fare' and 'Pclass' are related to survival; multicollinearity may exist.
6. Titles (Mr/Mrs/Miss) give additional information — e.g., 'Mrs' and 'Miss' have higher survival.

Recommended next steps:

- Perform feature engineering (combine SibSp & Parch into family size, encode titles, fill NA smartly).
- Train predictive models (Logistic Regression, Random Forest) to predict survival.
- Use cross-validation and address class imbalance if building models.